THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 14: **Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A cylinder of radius 14 cm and height 9 cm is melted and recast into cones of the same radius and height 3 cm. The number of cones is: 1

(a) 18 **(b)** 9 **(c)** 8 **(d)** 10

2. A solid sphere of radius 12 cm is melted and recast into a solid cylinder of radius 6 cm. The height of the cylinder is: 1

(a) 64 cm **(b)** 63 cm **(c)** 66 cm **(d)** 65 cm

3. Take $\pi = \frac{22}{7}$. The volume of a cone of radius 10.5 cm and height 14 cm is: 1

(a) 1779 cm^3 **(b)** 1942 cm^3 **(c)** 1455 cm^3 **(d)** 1617 cm^3

4. Take $\pi = \frac{22}{7}$. The volume of a cylinder of radius 14 cm and height 43 cm is: 1

(a) 23839 cm^3 **(b)** 26488 cm^3 **(c)** 31787 cm^3 **(d)** 29137 cm^3

5. The diagonal of a cuboid measuring 6 cm $\times$ 8 cm $\times$ 24 cm is: 1

(a) 27 cm **(b)** 26 cm **(c)** 25 cm **(d)** 28 cm

6. Take $\pi = \frac{22}{7}$. A toy is a cone of height 9 cm placed on a hemisphere, both of radius 21 cm. Its total volume is: 1

(a) 23562 cm^3 **(b)** 25919 cm^3 **(c)** 21205 cm^3 **(d)** 28276 cm^3

7. A solid sphere of radius 3 cm is melted and recast into small solid spheres of radius 1 cm. The number of small spheres is: 1

(a) 24 **(b)** 36 **(c)** 30 **(d)** 27

8. A cuboid 175 cm $\times$ 35 cm $\times$ 20 cm is melted and recast into cubes of edge 5 cm. The number of cubes is: 1

(a) 987 **(b)** 982 **(c)** 980 **(d)** 978

9. **Assertion (A):** The volume of a sphere of radius 7 cm is $\frac{4312}{3}$ cm^3 (take $\pi = \frac{22}{7}$). 1

Reason (R): The volume of a sphere of radius r is $\frac{4}{3}\pi r^3$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** When a metal sphere is melted and recast into a cylinder, the volume of the metal stays the same. 1

Reason (R): The surface area of a sphere of radius r is $4\pi r^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Take $\pi = \frac{22}{7}$. A cylinder of height 12 cm has volume 1848 cm^3. Find its radius and its curved surface area. 2

12. A cuboid measures 12 cm × 24 cm × 24 cm. Find the length of its diagonal, its total surface area and its volume. 2

13. Take $\pi = \frac{22}{7}$. A toy is a cone of radius 21 cm and height 28 cm resting on a hemisphere of the same radius. Find its total volume and its total surface area (curved parts only). 2

14. Take $\pi = \frac{22}{7}$. A cylinder of height 22 cm has volume 13552 cm^3. Find its radius and its curved surface area. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A solid sphere of radius 12 cm is melted and recast into a solid cylinder of radius 4 cm. Find the height of the cylinder and its curved surface area (Take $\pi = \frac{22}{7}$). 3

16. Take $\pi = \frac{22}{7}$. Water from a full cylindrical tank of radius 14 cm and height 10 cm is poured into an empty rectangular tank with a base 11 cm × 7 cm. Find the height of water in the rectangular tank. 3

17. Take $\pi = \frac{22}{7}$. A hemispherical bowl has inner radius 10.5 cm. Find its capacity, and the number of such bowls that can be filled from a vessel holding 7276.5 cm^3 of soup. 3

OR

A solid sphere of radius 3 cm is melted and recast into a solid cylinder of radius 6 cm. Find the height of the cylinder and its curved surface area (Take $\pi = \frac{22}{7}$). 3

18. Take $\pi = \frac{22}{7}$. Water from a full cylindrical tank of radius 14 cm and height 15 cm is poured into an empty rectangular tank with a base 22 cm × 14 cm. Find the height of water in the rectangular tank. 3

OR

Take $\pi = \frac{22}{7}$. A hemispherical bowl has inner radius 21 cm. Find its capacity, and the number of such bowls that can be filled from a vessel holding 38808 cm^3 of soup. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A solid metallic cone has a base radius of 7 cm and a vertical height of 24 cm. Take $\pi = \frac{22}{7}$. 5

(i) Find its slant height.
(ii) Find its curved surface area.
(iii) Find its volume.
(iv) The cone is melted and recast into a solid cylinder of base radius 7 cm. Find the height of the cylinder.

20. Take $\pi = \frac{22}{7}$. An ice-cream cone of radius 42 cm and height 56 cm is completely filled and topped with a hemisphere of ice-cream of the same radius. Find the total volume of ice-cream and the volume needed for 36 such cones. 5

OR

A solid sphere of radius 6 cm is melted and 5 identical solid cylinders of radius 2 cm are made from the metal. Find the height of each cylinder. 5

END OF QUESTION PAPER — Total: 40 Marks